Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering *CGPA: 9.05*

Projects

Negotiate, Then Solve: LLM Agents for Timetabling | *Python, FastAPI, React, TypeScript, Gemini, RAG*

- Built an LLM agent that turns free-text requests from email, chat and a web portal into typed constraints and negotiates conflicts with the people affected through typed tool calls.
- Shipped it end to end as a React 19 + FastAPI app with role-based access and an approval queue, backed by **215 pytest tests**.
- Added hybrid BM25 + dense RAG over policy documents (**93.4%** Recall@5) and guardrails (authority checks, cited rules, human approval) under which **no attack succeeded** in 50 prompt-injection and spoofing tests.
- LoRA-tuned a 4B model on Gemini agent traces that beat its teacher (17/18 vs 15/18), then passed **190/190** scenarios.

IndicLLM: Transformer Pretraining & Reasoning SFT | *Python, PyTorch, ONNX, FastAPI, Docker*

- Pretrained four 25M-parameter Transformer LMs from PyTorch primitives on 500M tokens each, then fine-tuned them for multi-hop reasoning over 44 runs, reaching **0.96** accuracy vs a 0.48 baseline.
- Deployed all 10 models to run in the browser via ONNX Runtime Web with a KV cache (logits within 1e-4 of PyTorch), served by a Dockerized FastAPI app on Railway.

Peer-to-Peer Distributed File Sharing System | *C++, Multithreading, Sockets, OpenSSL, Linux*

- Built a BitTorrent-style file-sharing system in C++ and cut 1GB single-peer downloads from **193s to under 13s** by replacing a polling scheduler with self-scheduling worker threads.
- Added Primary-Backup tracker failover with race-free leader election, verified by **46 unit tests** under ASan/UBSan.

Warehouse Traffic Control: Distributed Robot Coordination | *JavaScript, Three.js, Web Workers, Vite*

- Built a live 3D browser simulator of a 200-robot warehouse in Three.js, deployed to GitHub Pages via GitHub Actions.
- Designed a lease-and-fencing protocol across 16 region managers with **0 collisions over 416K moves** under 40% message loss, ±20% clock drift and injected crashes.
- Added congestion-aware A* routing driven by gossiped region loads, raising tasks completed by **98%** at 150 robots.

Technical Skills

Languages: Python, JavaScript, TypeScript, C/C++, SQL, HTML/CSS, Bash.
Web & Backend: React, Next.js, Node.js, FastAPI, REST APIs, PostgreSQL, SQLite, Prisma ORM, Three.js.
GenAI & LLMs: LLM Agents, Tool Calling, RAG, LLM Guardrails, Fine-Tuning (LoRA), Gemini API, PyTorch.
Cloud, DevOps & Tools: Git, GitHub Actions CI/CD, Docker, Vercel, Railway, pytest, Linux.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd by GPA in the B.Tech CSE batch at IIIT Ranchi.
- **PGEE 2025:** Secured All-India Rank 100 in the IIIT Hyderabad Post Graduate Entrance Examination.
- **Megathon 2025:** Placed 3rd in the mobile computing track.

Interests

- Web novels and manga, chess and badminton.